

Multiple Choice (10 marks)

1. Jake and Molly each have a few different objects. They want to see which objects can float. How can they best compare the results of their investigations with each other?
 - (a) Drop each object in different liquids.
 - (b) Record in a table how each object behaves.
 - (c) Measure the length and mass of each object.
 - (d) Test the objects in groups from smallest to largest.
2. What most likely happens when winter and summer temperatures are below normal and precipitation rates are above normal?
 - (a) Meltwater decreases, and the size of the glacier decreases.
 - (b) Meltwater decreases, and the size of the glacier increases.
 - (c) Meltwater increases, and the size of the glacier decreases.
 - (d) Meltwater increases, and the size of the glacier increases.
3. Which part of a pine tree makes food?
 - (a) root
 - (b) cone
 - (c) trunk
 - (d) needle
4. Which statement about the Sun is true?
 - (a) The Sun is a very large star that exists far from the Milky Way Galaxy.
 - (b) The Sun is at the center of the Milky Way Galaxy.
 - (c) The Sun is a medium-sized star near the edge of the Milky Way Galaxy.
 - (d) The Sun has the Milky Way and several other galaxies moving in orbits around it.
5. Earth's core consists primarily of iron. Which observation about another planet would provide some evidence that the planet has an iron core?
 - (a) the presence of a fluid mantle
 - (b) the presence of Van Allen belts
 - (c) the presence of gravitational force
 - (d) the presence of water in three states

True / False (10 marks)

Answer True or False

6. True or False: Lipids bond together to create carbohydrates in living organisms.
7. True or False: In a solid material, particles are closely packed and remain in fixed positions, similar to spectators seated in a stadium.
8. True or False: The momentum of a 0.15 kg baseball moving at 40 m/s is $3.8 \text{ kg} \cdot \text{m/s}$.
9. True or False: A population includes many species in an area, while a community consists of only one species worldwide.
10. True or False: An atom is the smallest particle that can exhibit the properties of a compound.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how volcanic eruptions in Earth's early history contributed to the formation of the planet's structure, focusing on the materials that originated from these eruptions.
12. Discuss how different materials affect the speed of sound. Which material do you think allows sound to travel the fastest, and why? Provide examples to support your answer.

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